

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence and Data Science) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTAIC402 - DATABASE MANAGEMENT SYSTEM

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 Which of the following best defines a Database?
 - A) A collection of unrelated files
 - B) A collection of data organized for easy access, management, and updating
 - C) A single table of data
 - D) Only text-based data storage
- 2 Which of the following is an example of a DML (Data Manipulation Language) command?
 - A) CREATE
 - B) INSERT
 - C) GRANT
 - D) COMMIT
- 3 The CAP Theorem states that in distributed systems, it is impossible to simultaneously guarantee:
 - A) Consistency, Availability, and Partition Tolerance
 - B) Completeness, Accuracy, and Portability
 - C) Cache Management, Accessibility, and Performance
 - D) None of the above
- 4 In a columnar database like Apache Cassandra, data is stored:
 - A) Row-wise
 - B) Column-wise
 - C) In flat files
 - D) In XML format
- 5 A major weakness of RDBMS when compared to NoSQL systems is:
 - A) Poor support for transactions
 - B) Limited data integrity enforcement
 - C) Inability to handle unstructured or semi-structured data efficiently
 - D) Lack of security features
- 6 Which of the following best describes a Graph Traversal operation?
 - A) Comparing two flat files
 - B) Sequentially reading columns
 - C) Exploring connected nodes through relationships
 - D) Fetching records using SQL aggregate functions
- 7 In future database trends, Disruptive Database Technologies aim to:
 - A) Completely remove the use of SQL
 - B) Maintain legacy systems only
 - C) Improve scalability, flexibility, and handle diverse data models
 - D) Restrict database use to only structured data

- 8 APIs that 'Return SQL' are designed to:
- A) Convert NoSQL queries into relational SQL results
 - B) Only support graph operations
 - C) Enhance database security
 - D) Replace SQL entirely with proprietary commands
- 9 Which of the following is TRUE about Document Databases like MongoDB?
- A) They only support numeric data
 - B) They store data in key-value pairs
 - C) They store semi-structured data in JSON-like documents
 - D) They require strict table structures
- 10 Which of the following is an example of a Key-Value NoSQL database?
- A) MySQL
 - B) Oracle
 - C) Redis
 - D) PostgreSQL
- 11 A relation is said to be in First Normal Form (1NF) if:
- A) It has no transitive dependencies
 - B) It contains atomic (indivisible) values only
 - C) It has a composite primary key
 - D) All attributes are foreign keys
- 12 In the Three-Level Database Architecture, the level closest to the end user is:
- A) Internal Level
 - B) Conceptual Level
 - C) External Level
 - D) Physical Level

Q2. Solve the following.

- A) List five responsibilities of a database management system. For each responsibility, explain the problems that would arise if the responsibility were not discharged
- B) Define the following terms with examples:
- a) Data
 - b) Database
- Explain the difference between two-tier and three-tier architectures. Which is better suited for Web applications? Why?

Q3. Solve the following.

- A) Consider the foreign key constraint from the *dept_name* attribute of the *instructor* to the *department* relation. Give examples of inserts and deletes to these relations, which can cause a violation of the foreign key constraint
- B) Differentiate between Instance and Schema in the context of databases. Consider the relational database given below:
- employee(person name, street, city)
- works(person name, company name, salary)
- company(company name, city)
- Give an expression in the relational algebra to express each of the following queries:
- a. Find the names of all employees who live in "Miami".
 - b. Find the names of all employees whose salary is greater than \$100,000.
 - c. Find the names of all employees who live in "Miami" and whose salary is greater than \$100,000.

Q4. Solve Any Two of the following.

- A) Define NoSQL. What does the term stand for, and how is it different from SQL? Explain the disadvantages of NoSQL technology with respect to consistency, query complexity, or lack of standardization.

- B) Discuss the disadvantages of NoSQL technology in terms of scalability, security, 6 or standardization. Describe the CAP theorem and briefly explain its three components.
- C) What is a Key-Value database? Provide two examples of Key-Value stores. 6 Design a simple schema using a Key-Value store for managing user profiles in a mobile application. Explain your design choices.

Q5. Solve Any Two of the following.

A) Define Document Database. Provide two examples of popular Document Database systems. Show with an example how to insert and retrieve data from a Document Database using MongoDB.

B) Explain the working of the query given below. How does MongoDB store this 6 document, and what advantages does this structure provide over traditional relational databases, particularly for applications with dynamic or evolving schemas?

```
db.users.insertOne({
  user_id: 501,
  name: "Rahul Sharma",
  email: "rahul.sharma@example.com",
  age: 30,
  country: "India",
  interests: ["travel", "reading", "technology"]
});
```

List any three key characteristics of Apache Cassandra that make it suitable for 6 distributed environments. Design a MongoDB collection structure to store product reviews with the following fields:

Review ID, Product ID, User ID, Rating (1-5), and Review Text.

Write the MongoDB insertOne() command to add a review and the find() command to retrieve all reviews for a specific product.

Q6. Solve Any Two of the following.

A) List different types of graph data structures used in graph databases. Explain how 6 SQL, NoSQL, and Graph databases differ in terms of data structure, storage approach, and system performance.

B) List the different graph data structures and explain their significance with 6 examples. A graph database contains the following relationships:

User 'A' follows 'B', 'B' follows 'C', 'C' follows 'D', 'D' follows 'A'.

Draw the graph and discuss possible graph traversal outcomes starting from 'A'.

C) With the help of the following schema, illustrate how NEO4J is used for storing 6 and querying relationship-oriented data. Explain the structure of nodes, relationships, and demonstrate graph traversal using Cypher queries.

*** End ***